

2.3.3 Number of Repetitions

دہرائی (لوپ) کارکردگی کا سب سے بڑا عامل ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 11 of 24

2.3.3 Number of Repetitions

LEARN IT

Repetitions (loops) are the most significant factor in algorithm efficiency. The **structure of the loop** decides how time grows with n .

Type of repetition	Conceptual impact	Efficiency grade
Linear loop	Processes each element exactly once. Time grows in proportion to size.	Good — $O(n)$
Logarithmic repetition	Reduces the problem size by half each iteration (binary search). Huge time savings.	Excellent — $O(\log n)$
Nested loops	For each element, process all other elements. Operations explode.	Poor — $O(n^2)$

WORKED EXAMPLE

Worked example: count the inner-body runs.

```
n = 5
linear = 0
for i in range(n):
    linear = linear + 1

nested = 0
for i in range(n):
    for j in range(n):
        nested = nested + 1

print(linear, nested)
```

OUTPUT
5 25

One loop: 5. Nested loops: $25 = 5^2$. For $n = 1,000$ that would be 1,000 versus 1,000,000.

REMEMBER IT

Find the **deepest loop**. One loop $\approx O(n)$. Loop inside a loop $\approx O(n^2)$. A loop that halves $n \approx O(\log n)$.

COMMON MISTAKE

Mistake: counting the word `for` and ignoring how the index moves.

Fix: a loop `while n > 1: n = n // 2` is logarithmic, even though it is still “a loop”.

دہرائی (لوپ) کارکردگی کا سب سے بڑا عامل ہے۔ ایک بار ہر عنصر پر چلنے والا لوپ $O(n)$ (اچھا)، ہر بار مسئلہ آدھا کرنے والا $O(\log n)$ (بہترین)، اور اندرونی لوپ $O(n^2)$ (خراب) ہے۔ اندرونی جسم کے چلنے کی گنتی سے یہ فرق صاف نظر آتا ہے۔

CHECK YOURSELF

1. What grade does the source give nested loops?
2. Why is binary search's loop logarithmic rather than linear?
3. If $n = 5$, how many times does a fully nested n -by- n pair of loops run?

Answers: 1. Poor — $O(n^2)$. 2. Each iteration halves the remaining problem, so the number of iterations is about $\log n$. 3. 25.

Next lecture: 2.3.4 The Efficiency Evaluation Framework